

x	y	z	$x+y+z$	$(x+y+z)'$	x'	y'	z'	$x' \cdot y' \cdot z'$
0	0	0	0	1	1	1	1	1
0	0	1	1	0	1	1	0	0
0	1	0	1	0	1	0	1	0
0	1	1	1	0	1	0	0	0
1	0	0	1	0	0	1	1	0
1	0	1	1	0	0	1	0	0
1	1	0	1	0	0	0	1	0
1	1	1	1	0	0	0	0	0

x	y	z	$x \cdot y \cdot z$	$(x \cdot y \cdot z)'$	x'	y'	z'	$x' + y' + z'$
0	0	0	0	1	1	1	1	1
0	0	1	0	1	1	1	0	1
0	1	0	0	1	1	0	1	1
0	1	1	0	1	1	0	0	1
1	0	0	0	1	0	1	1	1
1	0	1	0	1	0	1	0	1
1	1	0	0	1	0	0	1	1
1	1	1	1	0	0	0	0	0

(c) The distributive law of + over $\cdot$.

x	y	z	$y \cdot z$	$x + (y \cdot z)$	$(x+y) \cdot (x+z)$	$(x+y) \cdot (x+z)$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	1	1	1
1	0	0	0	1	1	1
1	0	1	0	1	1	1
1	1	0	0	1	1	1
1	1	1	1	1	1	1

** 2-5: Simplify the following Boolean functions to a minimum number of literals.

- (a) $xy + xy'$ (b) $(x+y)(x+y')$ (c) $xyz + x' + xy'z'$ (d) $z + zx'y$ (e) $(A+B)(A+B')$

Solution: (a) $xy + xy' = x \cdot (y + y')$ (Postulate 4(a))
 $= x \cdot 1$ (Postulate 5(a))
 $= x$ (Postulate 2(b))

(b) $(x+y)(x+y') = x + yy'$ (Postulate 4(b))
 $= x + 0$ (Postulate 5(b))
 $= x$ (Postulate 2(a))